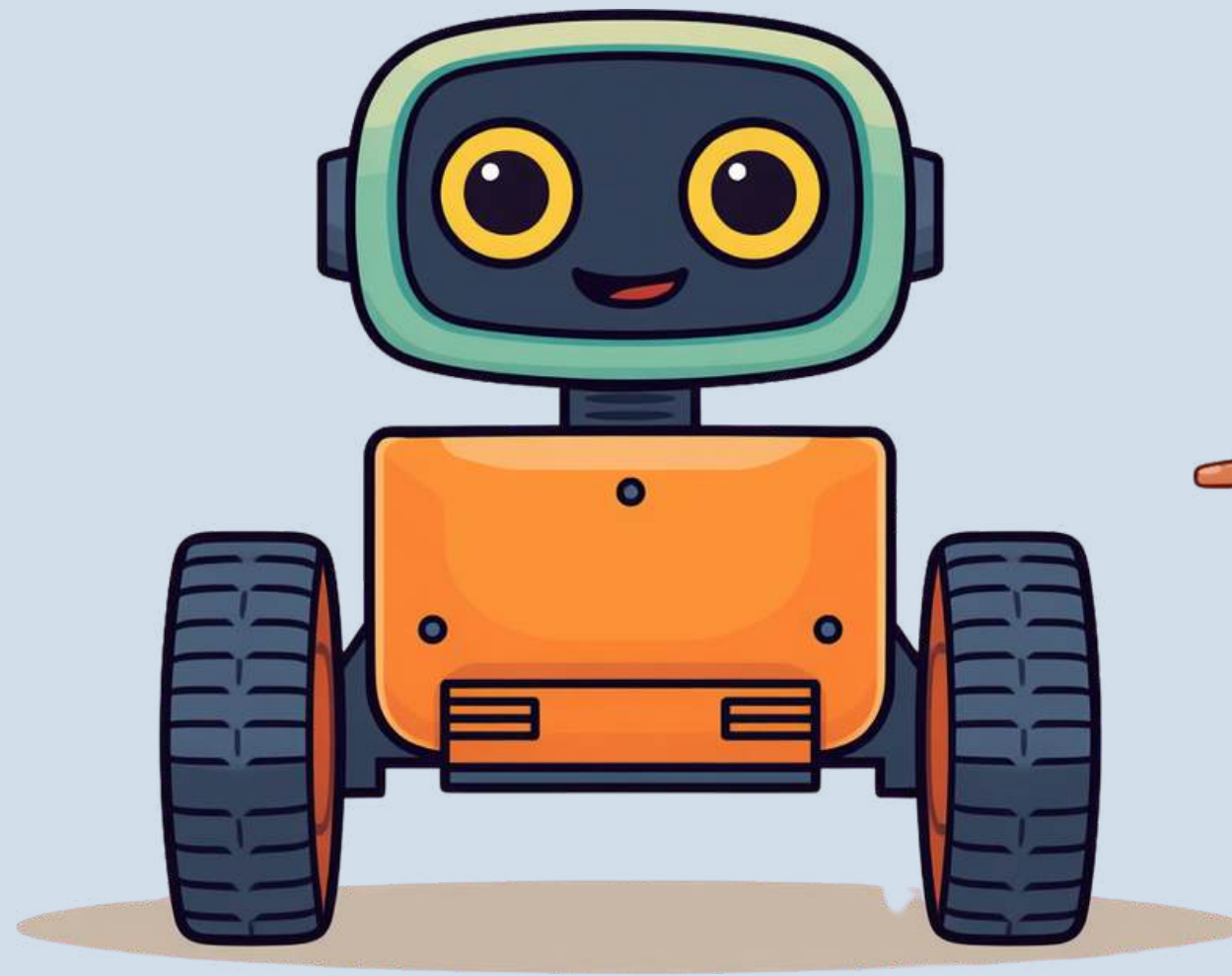


**ARDUINO — SMART OBSTACLE-
AVOIDING ROVER
EXPLORE ROBOTICS & SENSORS
WITH YOUR MINI ROBOT!**



HOW DOES A ROBOT SEE & DECIDE?

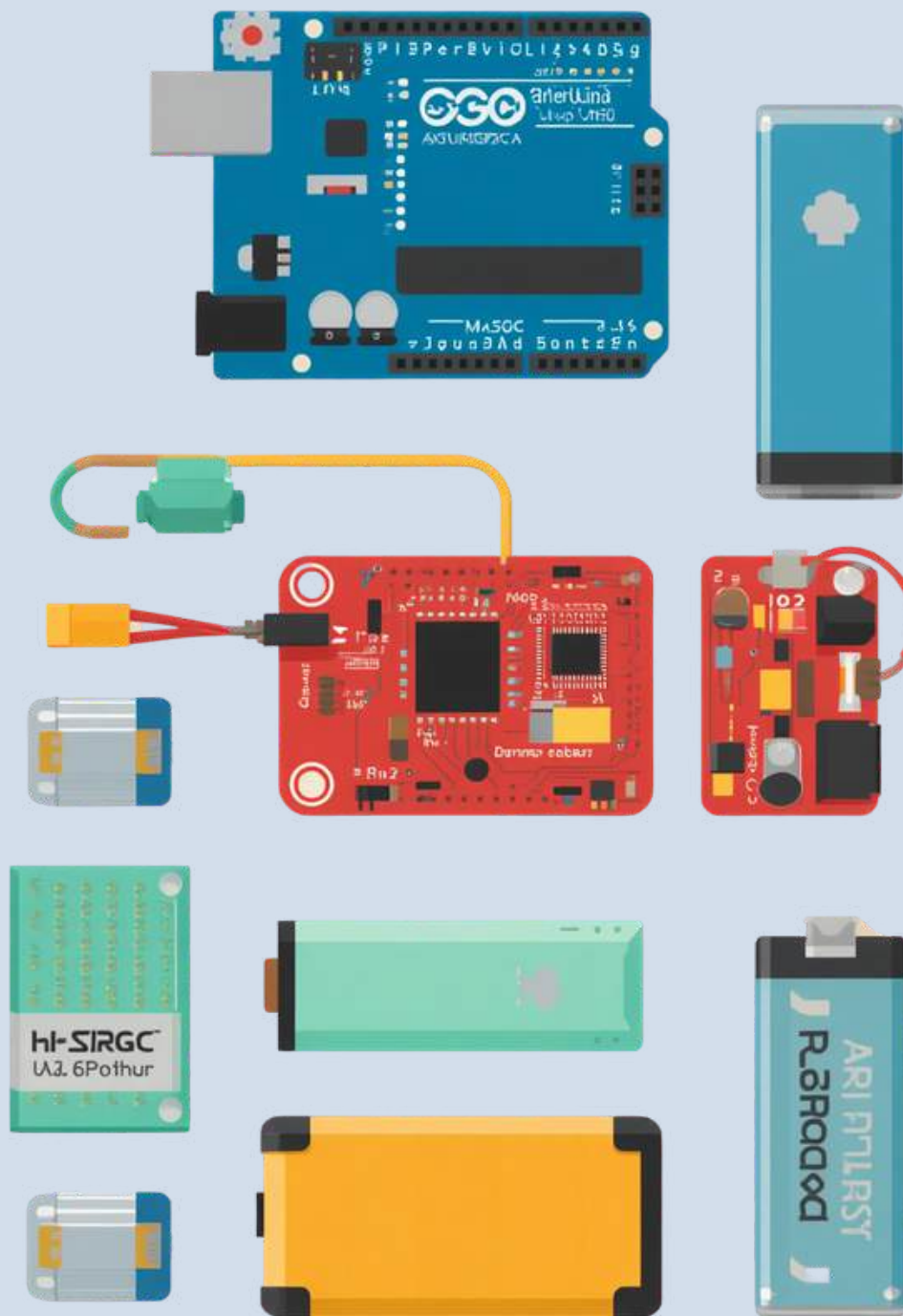


How does Pragyan avoid crashing into moon rocks without a driver?

🦇 Ultrasonic sensor = bat's echo in Western Ghats!

🚦 Arduino = traffic police at busy Mumbai crossroads!
Batao kids, what do you think the rover uses to 'see' obstacles?

MEET YOUR ROBOT'S PARTS!



- 🧠 **Arduino UNO — the brain**
- 👁️ **HC-SR04 — the bat's eyes**
- 🚗 **L298N Motor Driver — the auto-rickshaw driver's accelerator**

Wiring Callouts:

- **Trig pin → Arduino pin 9**
- **Echo pin → Arduino pin 10**
- **Motor pins → Arduino 5, 6, 7, 8**

HOW DOES THE ROBOT DECIDE? LET'S CODE!

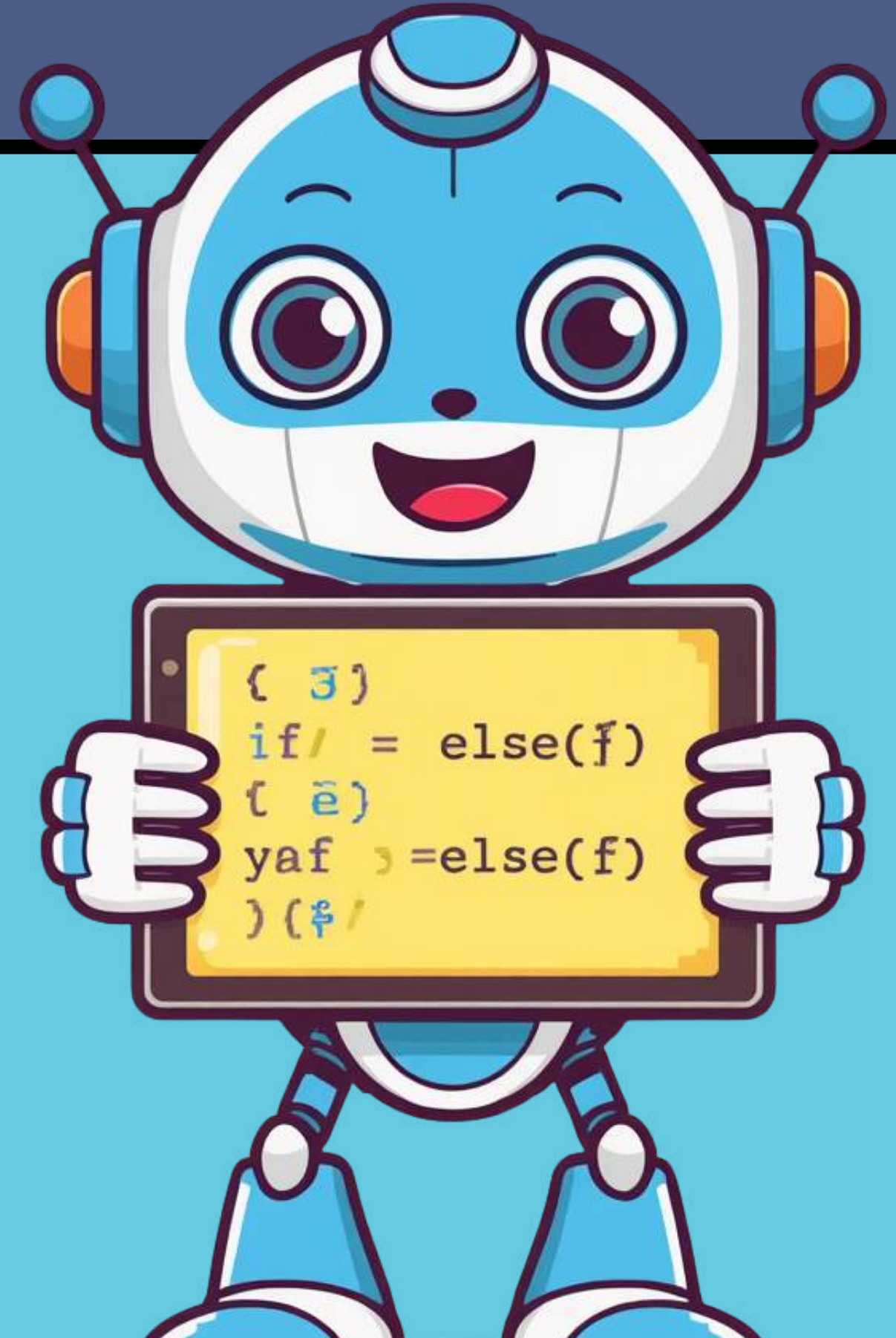
🔊 The rover sends a sound wave & waits for the echo — like clapping in a Rajasthan haveli!

📏 $\text{distance} = (\text{pulseDuration} \times \text{speed of sound}) \div 2$

🚦 if distance < 15 cm:
→ STOP & turn right!

✅ else:
→ Move forward!

💡 pulseDuration = time echo took to come back



```
{ 3 }  
if/ = else(f)  
{ e }  
yaf >=else(f)  
{ f/
```

LET'S SEE IT IN ACTION! LIVE CODING & QUESTION

 Watch the code come alive!

 Key lines to watch:

- Trig pin → Pin 9 sends the wave
- Echo pin → Pin 10 listens back
- if (distance < 15) → stop & turn!

 Think About It:

"What will happen if we change 15 cm to 2 cm? Batao!"

 Too short (2 cm) → Rover crashes before stopping!

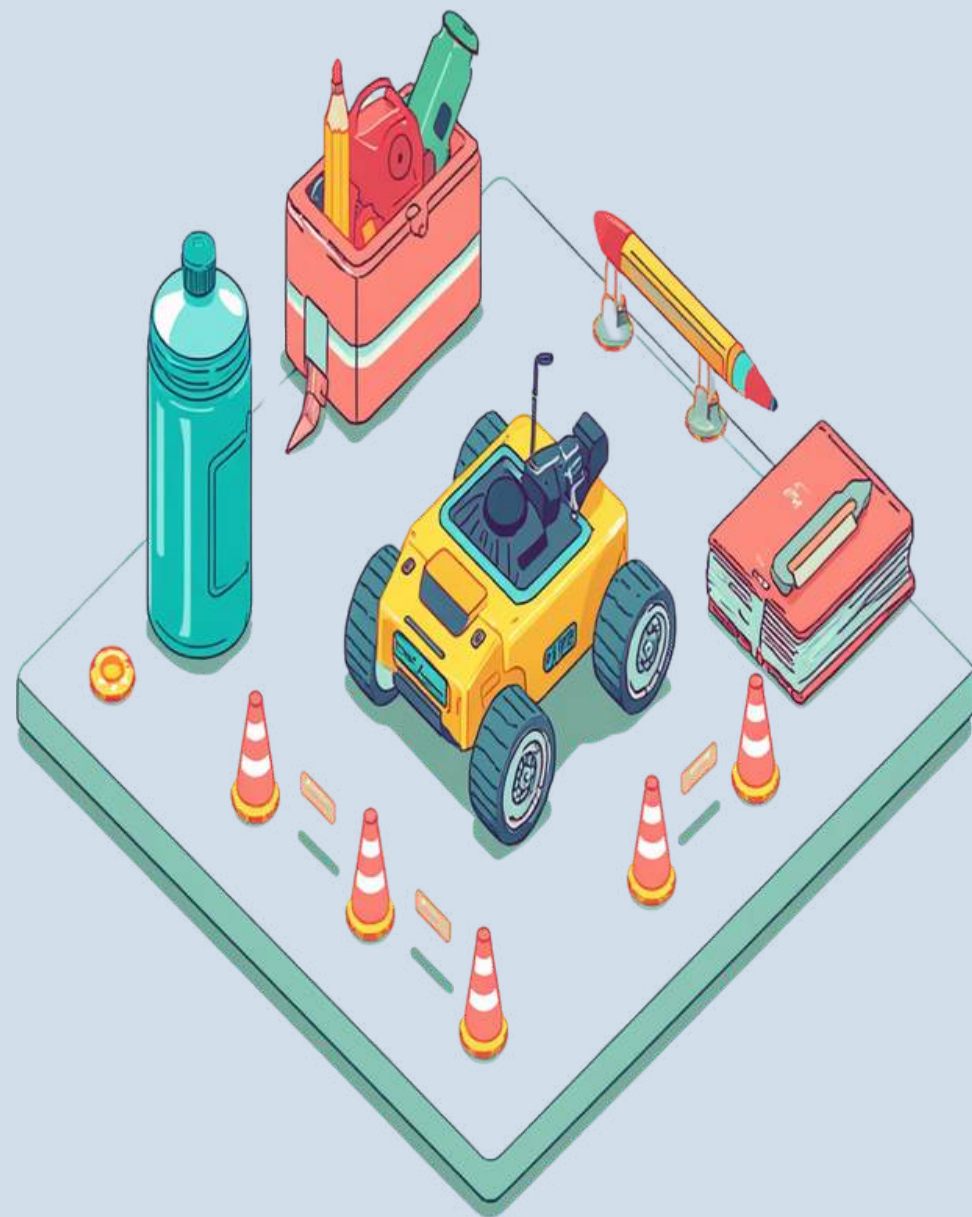
 Just right (15 cm) → Safe stopping distance!

 Too long (50 cm) → Rover stops too early!



```
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d  
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umitten (<=15)  
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time}  
=>
```

TESTING & TROUBLESHOOTING YOUR ROVER!



- 🔧 If motors run backward, switch motor wires!
- 📡 If distance reads zero, check sensor wiring & battery.
- 🔋 Safety: Always check battery terminals before connecting!
- 💡 Jugaad Tip: Use your creative skills to fix small problems!

QUICK QUIZ: SUPER TINKERER CHALLENGE!

1. What part is the rover's brain?

a) Motor Driver b) Arduino UNO ✓ c) Ultrasonic Sensor

2. What does the HC-SR04 sensor do?

a) Moves motors b) Measures distance ✓ c) Powers the robot

3. If an object is closer than 15 cm, the rover will:

a) Go forward b) Stop and turn ✓ c) Speed up

You're a Super Tinkerer! ✨



THANK YOU SEE YOU IN NEXT SESSION !

 **Homework: Draw your own obstacle course at home and note how your rover would move!**

 **Keep tinkering, keep exploring!**

